1 Half of the chord length $= 24 \div 2 = 12$ **[M1]**

Uses Pythagoras' theorem: $r^2 = 5^2 + 12^2$ **[M1]**

13 **[A1]**

[Total Marks: 3]

2 Uses Pythagoras' theorem: $(8 - -4)^2 + (1 - 6)^2$ or $12^2 + (-5)^2$ **[M1]**

$144 + 25 = 169$ **[M1]**

13 **[A1]**

[Total Marks: 3]

3 $\Delta x = 4 - (-2) = 6$ and $\Delta y = 9 - 1 = 8$ **[M1]**

$\sqrt{6^2 + 8^2}$ **[M1]**

10 **[A1]**

[Total Marks: 3]

4 Use of Pythagoras' theorem: $(k - 1)^2 + (19 - 4)^2 = 17^2$ **[M1]**

$(k - 1)^2 + 15^2 = 289$ or $(k - 1)^2 = 64$ **[M1]**

$k - 1 = 8$ and $k - 1 = -8$ (or $k^2 - 2k - 63 = 0$

(k-9)(k+7) = 0)[$M1$]

9 and -7 **[A1]**

[Total Marks: 4]

5 (a) $(x - 7)^2 + x^2 = (x + 1)^2$ **[M1]**

$x^2 - 14x + 49 + x^2 = x^2 + 2x + 1$ **[M1]**

$2x^2 - 14x + 49 = x^2 + 2x + 1$

$x^2 - 16x + 48 = 0$ **[A1]**

(b) $(x - 4)(x - 12) = 0$ **[M1]**

$x = 4$ or $x = 12$ **[A1]**

Rejects $x = 4$ because $x - 7$ would be -3 (length must be positive), so $x = 12$ **[B1]**

Perimeter $= 5 + 12 + 13 = 30$ **[A1]**

[Total Marks: 7]

6 (a) $AM = 16 \div 2 = 8$ **[M1]**

$PM = \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ with no errors seen **[A1]**

(b) In triangle AMQ, $QM^2 = 17^2 - 8^2$ or $289 - 64$ **[M1]**

$QM = \sqrt{225} = 15$ **[A1]**

$PQ = 6 + 15 = 21$ **[A1]**

[Total Marks: 5]

7 (a) $(2x+1)^2 = (x-1)^2 + (x+6)^2$ **[M1]**

$4x^2 + 4x + 1 = (x^2 - 2x + 1) + (x^2 + 12x + 36)$ **[M1]**

$4x^2 + 4x + 1 = 2x^2 + 10x + 37$ **[M1]**

$2x^2 - 6x - 36 = 0$

$x^2 - 3x - 18 = 0$ with no errors seen [A1]

(b) $(x-6)(x+3) = 0$ **[M1]**

$x = 6$ or $x = -3$ **[A1]**

Selects $x = 6$ and rejects $x = -3$ because lengths must be strictly positive **[B1]**

(c) Base $= 6 - 1 = 5$ and Height $= 6 + 6 = 12$ **[M1]**

Area $= \frac{1}{2} \times 5 \times 12$ **[M1]**

30 **[A1]**

[Total Marks: 10]

8 Let M be the midpoint of AB. $AM^2 = 8.5^2 - 4^2$ **[M1]**

$AM = \sqrt{72.25 - 16} = \sqrt{56.25} = 7.5$ **[M1]**

Chord $AB = 2 \times 7.5 = 15$ **[A1]**

[Total Marks: 3]

9 (a) $AM = 15 \div 2 = 7.5$ **[B1]**

$d^2 + 7.5^2 = 8.5^2$ **[M1]**

$d = \sqrt{72.25 - 56.25} = \sqrt{16} = 4$ **[A1]**

(b) Let half-chord be x. $x^2 + 7.5^2 = 8.5^2$ **[M1]**

$x = \sqrt{72.25 - 56.25} = 4$ **[A1]**

Length of chord $= 2 \times 4 = 8$ **[A1]**

[Total Marks: 6]

10 (a) Let half the chord length be x. Uses $12.4^2 = x^2 + 5.2^2$ **[M1]**

$x = \sqrt{12.4^2 - 5.2^2}$ or $x = 11.257\ldots$ **[M1]**

22.5 (or 22.51 to 22.52) **[A1]**

(b) Let perpendicular distance from O to CD be y. Uses $12.4^2 = y^2 + 9^2$ **[M1]**

$y = \sqrt{12.4^2 - 9^2}$ or $y = 8.5299\ldots$ **[A1]**

Total distance $= 5.2 + 8.5299\ldots$ **[M1]**

13.7 (or 13.72 to 13.73) **[A1]**

[Total Marks: 7]

11 **(a)** Uses distance formula: $\sqrt{(6-(-2))^2 + (y-(-1))^2} = 10$ **[M1]**

$64 + (y+1)^2 = 100$ **[M1]**

$(y+1)^2 = 36$

$y + 1 = 6$ (since $y > 0$)

5 **[A1]**

(b) Let C be $(x, 0)$.

$AC^2 = (x-(-2))^2 + (0-(-1))^2 = (x+2)^2 + 1$ **[M1]**

$BC^2 = (x-6)^2 + (0-5)^2 = (x-6)^2 + 25$ **[M1]**

Equating distances squared: $x^2 + 4x + 5 = x^2 - 12x + 61$ **[M1]**

$16x = 56$

3.5 **[A1]**

[Total Marks: 7]

12 **(a)** $x^2 + (x+7)^2 = (x+8)^2$ **[M1]**

$x^2 + x^2 + 14x + 49 = x^2 + 16x + 64$ **[M1]**

$x^2 - 2x - 15 = 0$ **[A1]**

(b) $(x-5)(x+3) = 0$ or correct use of the quadratic formula **[M1]**

$x = 5$ (accept value 5 only, ignoring -3 if appropriately discarded) **[A1]**

(c) Base $= 5 + 7 = 12$ and Height $= 5$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 5 = 30$ **[A1]**

[Total Marks: 7]